

VirtualLadder: Using Interactive Projections for Agility Ladder Training

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ABSTRACT

Agility is the ability to change your body's position in a fast and efficient way while maintaining control of speed and direction. To develop this skill, the use of agility ladders is a widespread and well-known training method. While drills vary, having a human expert explaining and monitoring the exercises is usually advantageous. So far, only a few approaches to interactive systems for agility ladder training have been presented. In this work, we propose an interactive projection system to support an athlete in performing those drills. To investigate the effects of the location of those projections, we conducted an initial study with twelve participants using qualitative and quantitative methods. We found that while projecting instructions and feedback on the floor was favored, participants who were presented the same information in front of them on a projected screen partially performed better in a subsequent agility assessment.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI; • Applied computing → Interactive learning environments.

KEYWORDS

Sports, Interactive Training System, Agility, Projections

ACM Reference Format:

Felix Kosmalla, Fabian Hupperich, Anke Hirsch, Florian Daiber, and Antonio Krüger. 2021. VirtualLadder: Using Interactive Projections for Agility Ladder Training. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451638>

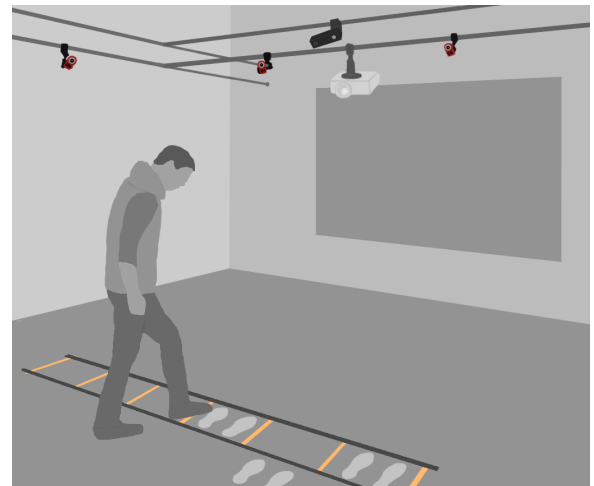


Figure 1: VirtualLadder: A physical ladder is augmented via a projection from the top. An optical tracking system is used to capture the position of the feet to give feedback and detect correct or incorrect steps. A second visualization projects the instructions to the wall in front of the participant.

1 INTRODUCTION

Agility and coordination are basic skills that every athlete should seek to improve [24]. These abilities are beneficial and are required in nearly every sport where athletes have to make quick decisions depending on their surroundings. While there are different definitions of agility (e.g [5, 31]), in the following we will define this

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CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan
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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00
<https://doi.org/10.1145/3411763.3451638>

term as the ability to change one's body's position in a fast and efficient manner while maintaining control of speed and direction. A common method to improve an athlete's agility and coordination skill is the use of a so called agility ladder. An agility ladder in its simplest form is a rope ladder that is placed on the ground (see Figure 1). The boxes between the rungs are used as stepping targets for a range of stepping sequences, varying in complexity and thus, providing drills for all skill levels [3]. These exercises are performed by the athlete stepping into the boxes of the agility ladder, following a sequence like the one seen in Figure 3. A sequence should be completed as fast as possible without making any mistakes (e.g. stepping in the wrong box or using the wrong foot). The most common way of learning those exercises in a training session is by an expert demonstrating the sequence. However, a trainer might not always be available because they are supporting other athletes, monitoring the group, or simply because the athlete is training alone. It is especially important for beginners that they perform the exercises with correct form and receive assistance in the event that they are experiencing difficulties during training [6]. Interactive training systems have shown to be able to, at least to some extent, fill this gap in other sports like slacklining [15], ballet [29], or general movement training [1]. To the best of our knowledge, the effects of interactive projections on training efficiency and user experience in interactive systems for agility ladder training have not been investigated in detail. In this work, we tested the effect of a visual feedback projected directly on an agility ladder on the floor as well as a projection in front of the user on a projection screen. With the presented system and study, we aim at giving initial insights in how such systems could be designed and evaluated in the future. The contribution of this work is the design and implementation of an interactive agility training system as well as an initial evaluation of the system. While this system was specifically tailored for agility ladder training, the concepts and insights provided in this work could inspire the design of applications that go beyond sports, e.g. rehabilitation systems or motor learning in general.

2 RELATED WORK

A classic approach to acquiring a new motor skill is observational learning: a novice watches an expert making a certain movement who may also give additional explanation about specific details [17]. Later on, the expert supervises the novice and gives individual feedback on points to improve. This paradigm was picked up by several interactive training systems. The investigated modalities range from on-body tactile feedback [9, 27], video feedback [1, 7, 30], and auditive guidance [20, 28] over to visual, projected feedback [25, 26] in different sport domains.

Projected In-Situ Feedback. The use of projections allows for using any surface as an output which leads to completely new design opportunities. While display based guidance systems might draw the attention of the user away from their interaction space, projections can bring guiding feedback closer to the action. In *SleeveAR* [26], projection is used to augment visual guidance directly onto the patient as well as on the floor during physical therapy exercises. Similarly, Sodhi et al. compared different visualizations projected directly on the participants hand to guide them through

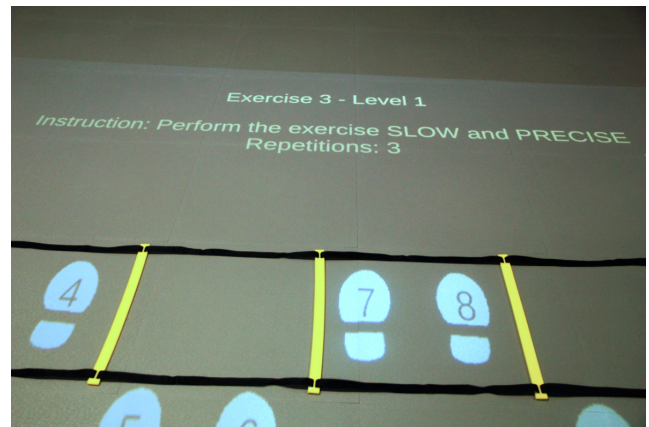


Figure 2: Right: The instructions for the first level projected directly above the ladder.

several movements [25]. Wiehr et al. [30] proposed a mobile, self-calibrating camera-projection unit which could be placed in front of an arbitrary bouldering wall to teach climbing movements. Similarly, the *Augmented Climbing Wall* by Kajastilla et al. [13] combines a regular climbing wall, a depth-camera, and a projector. Kosmalla et al. [14] investigated three different visualization methods to learn climbing movements by video demonstration. They concluded that a life-sized projection directly at the place of action (the climbing wall) was perceived to be the easiest to follow. Peres et al. [16] evaluated a projection system for gait training with crutches. These examples demonstrate the effectiveness of projected in-situ feedback to convey different aspects of learning a new motor skill.

Feedback Systems Using a Mirror Metaphor. While the works in the previous paragraph focus on feedback directly or close to the area of action, another approach is to present augmented feedback in front of the user, often using a mirror metaphor. In a form of augmented visual feedback, Anderson et al. [1] presented *YouMove*, an augmented reality mirror, teaching users dance movements via an augmented image of the trainee. By slowly reducing the amount of feedback during training, they avoided the development of a dependency on the system [21]. Kosmalla et al. [15] presented *Slackliner*, an interactive system for slackline training. Their system tracked and guided the user through their slackline training using real-time feedback. Hämäläinen [7] proposed a motion controlled video recording approach that presents the feedback on a screen in front of the user.

Influence of the Location of Feedback on Performance Improvement. While being presented feedback directly on the surface of interaction diminishes the complexity of matching the feedback to executed movements, there are numerous examples showing that adding further mental demand and visual guidance to sports training can increase the learning effect. A study by Roopchand-Martin et al. [18] examined the effects of a six-week traditional agility ladder program compared to dance video gaming. They used the *Illinois Agility Test* (IAT) [5] to assess the agility of the participants before and after the six-week training period. The results of their study showed an improvement in agility scores for both groups.

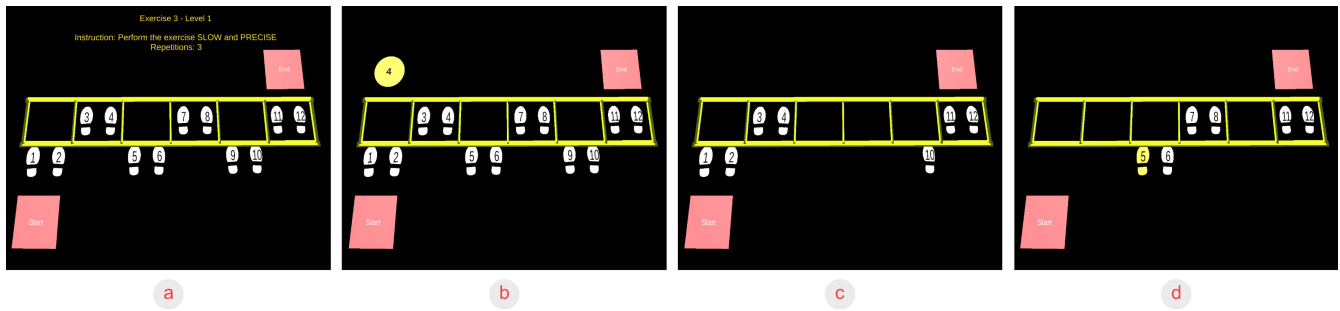


Figure 3: Overview of the user interface of the proposed system. a) The user is given a task description. b) After stepping on the start field, a countdown begins to indicate the start of the exercise. c) Over the course of three levels, the visual guidance in the form of footprints is reduced continuously up to the point of no clues at all. d) If the user makes a mistake (e.g. steps in the wrong field with the wrong foot), additional visual hints are revealed and the next correct step is highlighted in yellow.

Sano et al. [19] implemented an augmented ball game for training the receiving of a pass that visualized the trajectory and velocity of a soccer ball.

In past works, the location of the feedback was investigated. Sekhavat and Namani [23] compared two different visual feedback methods in gait rehabilitation training. They extended a treadmill with both a projector and a front mounted display which were used as feedback channel in each of the two conditions. The task of the participants was to step on virtual footsteps that were either augmented on the treadmills belt via a projection or in front of the participants on a display. Their work implies that for gait rehabilitation, augmented projection is more effective than the display in front of the participants.

Interactive Systems for Agility Ladder Training. There is only limited work investigating the use of interactive systems in agility ladder training. Boonim and Sanguansat presented *Automated Ladder*, a system that uses image processing to detect foot positioning of a person performing on an agility ladder [2]. While their system was able to determine the duration of an individual exercise, it did not feature error detection nor did it provide any guidance or feedback. A commercially available approach of an interactive agility trainer can be seen in the The Quick Board¹. The system combines a reactive mat that detects footsteps when walking on it and a tablet which displays instructions in front of the athlete at eye level. A study was conducted that indicated an increase in the athlete's agility and coordination skills after a four-week training program using the system [4]. While the user interaction in this systems also heavily relies on foot placement, the system proposed in this work allows for varied exercises spanning the whole length of a usual agility ladder.

3 DESIGN & IMPLEMENTATION

When designing the agility ladder trainer, we kept the classical training approach and concepts in mind [3]. Our setup consisted of a physical agility ladder placed on the floor within the tracking volume of a 12-camera optical tracking system (OptiTrack²). A Kinect v2 in combination with a projector was mounted at a height

of approximately 2.8m slightly offset the ladder on the ground. A projection screen was placed facing the participant, allowing either a projection directly on the floor or at eye level in front of the user (see Figure 1). We used the RoomAlive Toolkit [12] and Unity in combination with the Kinect for projection mapping on the floor. While the Kinect was used exclusively for projection mapping, the OptiTrack system was used to track the feet and interactions of the participants using reflective marker clusters on the shoes. Using the RoomAlive Toolkit provides the developer with a coarse 3D model of the to be projected on space that can be imported into Unity. After aligning the Unity coordinate system with the OptiTrack (i.e. physical world) coordinate system, arbitrary virtual objects can be placed in the virtual scene which are then projected in the real world at the correct position. Individual, virtual planes were placed within the boxes of the physical agility ladder. By attaching colliders to the planes, it was possible to register a step within those boxes.

Projection of the Visual Feedback. For our system we implemented two different visualizations. In the *Floor Projection* visualization, the visual feedback is directly projected on top of the agility ladder on the floor, matching the individual rungs of the ladder. As an alternative visualization, we implemented the *Wall Projection* which is displayed on a large screen in front of the athlete. To give a frame of reference, this visualization was extended by a virtual representation of the agility ladder as well as a representation of the athlete's feet which updated in real time. We chose these two types of visualization based on the related work outlined above.

As a visualization for guiding the athlete we chose numbered footprints which were projected either directly into the boxes of the physical agility ladder (see Figure 2) or included in the visualization of the *Wall Projection*. We chose this simple visualization over more complex feedback to allow the user's attention, being a limited resource, to be used to develop new movement skills [11].

Exercise Design. We designed the exercises based on three drill levels as recommended by Schreiner [22]. In the first level **Familiarization**, the goal is to perform the exercises and steps very precisely and with minimal to no errors if possible. Speed and quickness should be ignored in this phase, as it is more important to be precise rather than to be fast in the beginning. In the second level **Increase of Speed**, the athlete should increase the tempo

¹<https://www.thequickboard.com>

²<http://www.optitrack.com>

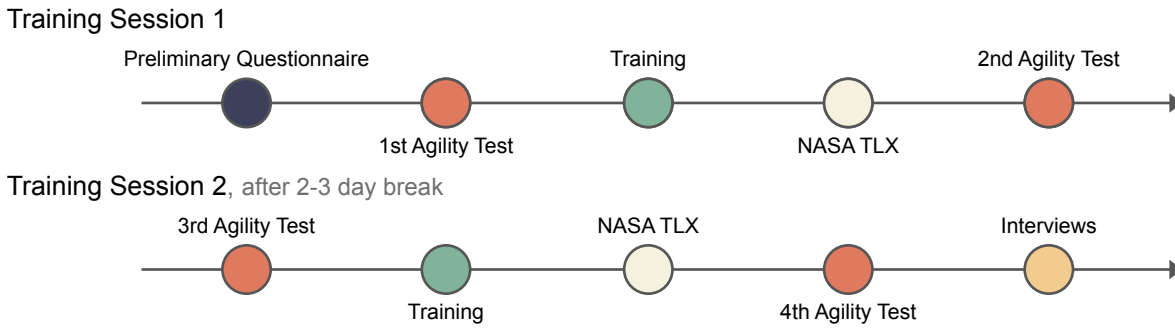


Figure 4: The overall procedure of the study sorted by the two training sessions and their order of execution.

while keeping precision. It is still important that the trainer (or, in this case, the system) corrects mistakes. The final **Training** level aims at reaching maximum speed while still being precise. This stage should only be considered when the athlete is able to perform the given footstep combination at a mid-paced tempo with almost little to no error.

We implemented the mechanics of the proposed system accordingly. The projections described above led the athlete through these levels for every exercise, including the correction of errors. As mentioned above, for each exercise the athlete undergoes three levels when performing a new stepping sequence. During the progression from the first to the third level, the quantity of feedback is reduced gradually while increasing the demand of speed. In the first level, the complete sequence is displayed and the athlete is asked to perform the sequence slowly but without any mistakes. To encourage learning, the number of visual footprints is reduced and the required speed is increased. In the final level, the system assumes that the athlete has memorized the stepping sequence and only gives assisting feedback when an error is registered. This is achieved by subsequent steps reappearing to help the athletes to reorient themselves. Additionally, the next step is highlighted (Figure 3 d). In doing so, the athlete should learn to complete the exercise on their own, which prevents developing a dependence on continuous feedback. The athlete needs to finish the sequences in this level in a set time (see Figure 3 c). A repetition is counted as success when the athlete did not make any mistakes and also stayed within the time limit. If the athlete cannot beat the given time or makes a mistake, an audio signal will notify them that they have failed the repetition and the run is not counted as a success. After completing this level, the whole exercise is completed. Each exercise in a training session introduces the athlete to a stepping sequence. Written instruction and further tips are displayed above the ladder (see Figure 3 a). This sequence has to be repeated three times for each level to complete the exercise. The correct step in a rectangle of the ladder is acknowledged by the footprint's disappearance and an audio signal.

4 USER STUDY

We conducted an initial between-subject user study to investigate the effects of the different visualization as described above. Our goal was to investigate whether there is a difference in improvement of the participants' agility based on the type of visualization and

which method involves more cognitive effort. The study has been approved by the ethical review board of the faculty of Mathematics and Computer Science at Saarland University.

4.1 Methods

4.1.1 Participants. Overall, twelve university students (two female, ($M=22.58$, $SD=1.62$)) participated in the study. Half of all participants had experience with coordination or agility training in general (e.g. soccer drills), three with agility ladder training in particular.

4.1.2 Design & Conditions. We tested two different visualization methods in a between-subject design: a) a projection directly on the floor on top of the agility ladder (the Floor Projection Group, (FPG)) and b) a projection on a big screen in front of them (the Wall Projection Group (WPG)).

4.1.3 Independent & Dependent Variables. The two conditions FPG and WPG were used as the two factor levels of the independent

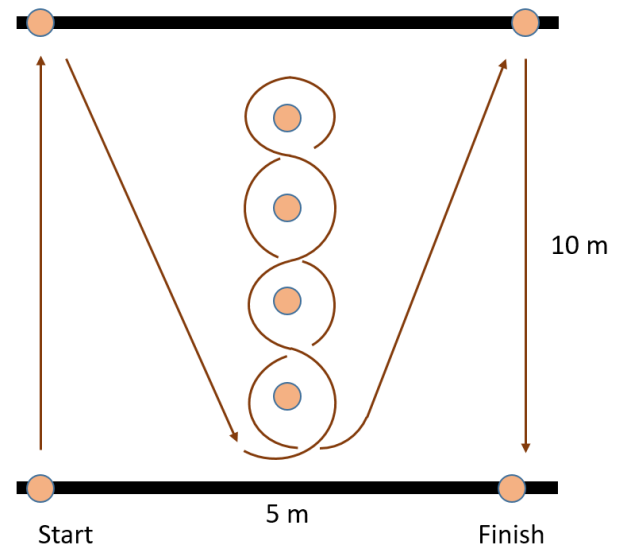


Figure 5: Schematic of the route of the IAT.

variable projection style. The dependent variable *agility* was measured with the Illinois Agility Test, *mental demand* and *frustration* were measured via the NASA TLX [8]. Furthermore, the number of failed repetitions in the final level of each exercise is also treated as a dependent variable. Finally, there was an assessment on how the participants described their experiences in the semi-structured interview.

4.1.4 Procedure. The participants were asked to attend two training sessions on two separate days (see study overview in Figure 4). After informed consent was obtained, demographics, previous experiences with agility ladder training, and coordination training in general was recorded. To assess the initial level of agility, the participants were asked to perform an IAT³ [5] after a warm up. A defined course around four cones (see Figure 5) had to be mastered as fast as possible, considering only the fastest out of three trials which is then recorded as the IAT score. The IAT was followed by a training session with the system, using either of the two conditions. A session with six different exercises, each repeated at least nine times was conducted. An initial introduction to the system and its interaction was given by the experimenter. Afterwards, only the system guided the participants through their training. During the training, the number of repetitions they needed in order to complete the last level of each exercise was recorded. In the last level, the participants had to perform the sequence in a set time for a total of three times, as precisely as possible (i.e. no stepping mistakes). If a participant did not beat the set time, they had to repeat the sequence, thus resulting in possible additional repetitions. After completing the first training session, the participants were asked to fill out a NASA TLX and then performed a second IAT following the same procedure as described above. The actual interaction time with the system was 30-35 minutes per training session. After a break of two to three days, the participants were invited back to the second training session which followed the same protocol as the first. While one single group of participants were only presented one type of visualization during the two training sessions, the participants had the chance to try out both visualizations at the end of the study. This allowed us to collect comparing qualitative feedback for both visualization in semi-structured interviews.

4.1.5 Hypotheses. Based on the related work outlined above ([18], [23], [26], respectively), we formulated the following hypotheses:

H₁: Training with the automated agility ladder training system improves the IAT score in the WPG to a higher degree than in the FPG.

H₂: There will be more overall failed repetitions in the WPG than in the FPG.

H₃: The NASA TLX shows that the FPG outperforms the WPG.

4.1.6 Apparatus. We used a lab setting and the system described above for the exercises with the interactive agility trainer. A gym in a neighboring building was used for the IAT, which was set up with cones as seen in Figure 6. In addition to the individual components of the system, we also employed a video camera for further analysis.



Figure 6: Actual setup of the IAT with cones marking the important turning points.

5 RESULTS

5.1 Quantitative Results

For the analysis, we used IBM SPSS Statistics (Version 26) predictive analytic software for Windows⁴. For all three hypotheses, we conducted an independent two-sample t-test. We tested one-sided because we used directed hypotheses. The requirements for an independent two-sample t-test were met. For all hypotheses, the Shapiro Wilk test showed a normal distribution of our data, $p > .05$, and the boxplots showed only at most two minimal outliers for each condition. We computed the differences of the IAT scores in the first training session (Test 1 & 2), second training session (Test 3 & 4) and between the first test of the first training session and the last test of the last training session (Test 1 & 4) (see Figure 4).

Regarding IAT, we could show a significant difference in the IAT scores during the first training session for the FPG ($M = -0.35$, $SD = 1.02$) and WPG ($M = -1.37$, $SD = 0.85$) conditions; $t(10) = 1.87$, $p < .05$, $d = 1.08$. The wall projection group showed a greater timely improvement from the first IAT measurement to the second. No significant difference in the IAT scores during the second training session for the FPG ($M = -0.18$, $SD = 0.46$) and WPG ($M = -1.37$, $SD = 0.85$) conditions; $t(10) = .87$, $p = .202$, $d = 0.50$. Furthermore, no significant difference in the IAT scores of the very first and last test for the FPG ($M = -0.73$, $SD = 0.78$) and WPG ($M = -1.49$, $SD = 0.1.08$) conditions; $t(10) = 1.38$, $p = .099$, $d = 0.79$.

The test for error rate revealed a significant difference in repetitions: the WPG ($M = 10.67$, $SD = 4.84$) had on average more failed repetitions than the FPG ($M = 4.50$, $SD = 2.07$); $t(6.77) = -2.87$, $p < .05$.

For H_3 , we used Bonferroni-Holm adjustment [10] to avoid alpha error accumulation because we had four t-tests here. This resulted in the following p value threshold to be undercut: 0.05, 0.025, 0.017, and 0.013. The results revealed only one t-test to be significant. The mental demand for the wall projection group was higher in the second training session, $t(10) = -2.13$, $p < .05$, $d = -1.23$. There was no significant difference between the mental demand or frustration

³We followed the test protocol described here <https://www.topendsports.com/testing/tests/illinois.htm>

⁴<https://www.ibm.com/products/spss-statistics>, last accessed: January 9th, 2021

in the first training session or between the frustration in the second training session between the two conditions which means this hypothesis could only be partially supported by our data.

5.2 Qualitative Results

After finishing both training sessions, a semi-structured interview was conducted. Nine out of the twelve participants agreed with the first questions that they can imagine training with the system in the future, given the availability of such a system at the local gym or at their sports club. Three of those nine participants criticized the lack of variety in the exercises. No specific reasons were provided by the three participants who did not like to continue training with the system other than general lack of interest in agility and coordination training. The second topic assessed whether the participants liked and enjoyed training with the system. All participants stated that they enjoyed the training or the experience of testing such a system. Three participants stated that they liked the possibility to train on their own without a real trainer. Lastly, the participants were asked what they did not like about training with the system. The focus switch between the projected screen in front of them and the ladder on the floor was mentioned as problematic by most of the participants in the WPG. Three participants saw room for improvement in the accuracy of the step detection. Some participants did not like the fixed order of the training session; they would have enjoyed skipping exercises that they already knew. In the final part of the interview, the participants were briefly introduced to the second visualization method and they were also given the opportunity to try it out. Then they were asked which of the two approaches they preferred. Eleven (five from FPG, six from WPG) out of the twelve participants stated they strongly preferred the floor projection. The overall consensus was that it is more intuitive as the natural direction of participants' gaze during the exercises is to the floor. Two participants who preferred the floor projection mentioned that the choice of guidance makes no difference for more experienced athletes.

6 DISCUSSION

According to the IAT results, only the participants in the WPG improved their agility and only in the first TS, thus H_1 could only be partially supported. As already indicated by Roopchard-Martin et al. [18] additional mental effort can lead to an increased training effect, which could explain the improvement of the WPG. While we could not find a significant improvement of the FPG, looking at the lower mental demand, this visualization could motivate users to continue their training, justifying its viability. The number of failed repetitions was significantly higher in the WPG, confirming H_3 . This is in agreement with the higher Mental Demand Scores for the WPG in the second training session, as well as with the feedback from the semi-structured interviews. Multiple participants stated that they found it substantially more difficult to translate the guidance on the wall into real motion as opposed to having the guidance directly where they have to step. Recovering from a mistake was perceived harder in the WPG due to being forced to look up and down from the screen to the floor. Mixed feedback was received from both systems, therefore preventing us from confirming that one guidance is better fitted than the other. Considering

a combination of both visualizations might be beneficial. Nearly all participants enjoyed the guidance projected on the floor, citing reasons such as it being more intuitive and easier to use. A reason for this seems to be the spatial closeness between guidance and foot positioning. The majority of the participants stated they would continue training, indicating that the system made the training enjoyable and interesting. Participants that did not want to continue training with the system said they wouldn't like to train for agility and coordination in general, no matter what training method would be used.

While the number of participants for this initial study was limited, we believe that this work could still inform future research of interactive training assistants. In addition, the introduction of a condition in which the participants train without the system described above as a baseline could be used to confirm whether the interactivity component is beneficial in general.

7 CONCLUSION & FUTURE WORK

In this work, an interactive agility ladder trainer was implemented and evaluated in an initial study ($N=12$), comparing two different feedback visualizations. The results of the IAT [5] showed that in the first training session there was a significant improvement in the agility skill level of the participants that trained with the wall projected guidance compared to those who trained with the floor projection. The finding that the wall projected guidance was scored more mentally challenging is also supported by the results of the semi-structured interviews. Eleven out of twelve participants stated they would like to continue training with the floor projected visualization. This visualization was perceived overall to be more intuitive and easier to use. Based on these results, the question which visualization method is better cannot be answered clearly. As a consequence, the goal for further research would be to experiment with the two guidance methods in order to investigate which aspects benefit the different stages of the exercise, the individual athletes, etc. Since the participants only had limited time training with the system, we cannot make any conclusions about long-term effects. We plan to investigate this in future, more extended studies.

ACKNOWLEDGMENTS

This work is partially supported by the German Federal Ministry of Education and Research (16SV8364).

REFERENCES

- [1] Fraser Anderson, Tovi Grossman, Justin Matejka, and George Fitzmaurice. 2013. YouMove: Enhancing Movement Training with an Augmented Reality Mirror. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST '13). ACM, New York, NY, USA, 311–320. <https://doi.org/10.1145/2501988.2502045>
- [2] K. Boonim and P. Sanguansat. 2018. Athletes Performance Evaluation by Automated Ladder for Speed and Agility. In *2018 15th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*. 106–109. <https://doi.org/10.1109/ECTICon.2018.8620013>
- [3] Lee Brown and Vance Ferrigno. 2014. *Training for speed, agility, and quickness*, 3E. Human Kinetics.
- [4] Andrew J. Galpin, Yuhua Li, Corey A. Lohnes, and Brian K. Schilling. 2008. A 4-Week Choice Foot Speed and Choice Reaction Training Program Improves Agility in Previously Non-Agility Trained, but Active Men and Women. *Journal of Strength and Conditioning Research* 22, 6 (2008), 1901–1907. <https://doi.org/10.1519/JSC.0b013e3181887e3f>
- [5] Bud Getchell. 1979. *Physical fitness: A way of life*. ERIC.

- [6] The Ultimate Agility Ladder Guide. accessed November, 2, 2019. www.greatmeltoncc.co.uk/news/wp-content/uploads/2011/02/Ladder-Drills-Agility-Balance-Coordination1.pdf. (accessed November, 2, 2019).
- [7] Perttu Hämäläinen. 2004. Interactive Video Mirrors for Sports Training. In *Proceedings of the Third Nordic Conference on Human-Computer Interaction* (Tampere, Finland) (NordiCHI '04). Association for Computing Machinery, New York, NY, USA, 199–202. <https://doi.org/10.1145/1028014.1028044>
- [8] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.
- [9] Mahmoud Hassan, Florian Daiber, Frederik Wiehr, Felix Kosmalla, and Antonio Krüger. 2017. FootStriker: An EMS-Based Foot Strike Assistant for Running. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 1, Article 2 (March 2017), 18 pages. <https://doi.org/10.1145/3053332>
- [10] Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian journal of statistics* (1979), 65–70. <https://doi.org/10.2307/4615733>
- [11] Katherine Isbister and Florian "Floyd" Mueller. 2015. Guidelines for the design of movement-based games and their relevance to HCI. *Human-Computer Interaction* 30, 3–4 (2015), 366–399.
- [12] Brett Jones, Rajinder Sodhi, Michael Murdock, Ravish Mehra, Hrvoje Benko, Andrew Wilson, Eyal Ofek, Blair MacIntyre, Nikunj Raghuvanshi, and Lior Shapira. 2014. RoomAlive: Magical Experiences Enabled by Scalable, Adaptive Projector-camera Units. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (Honolulu, Hawaii, USA) (UIST '14). ACM, New York, NY, USA, 637–644. <https://doi.org/10.1145/2642918.2647383>
- [13] Raine Kajastila, Leo Holsti, and Perttu Hämäläinen. 2016. The Augmented Climbing Wall: High-Exertion Proximity Interaction on a Wall-Sized Interactive Surface. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 758–769. <https://doi.org/10.1145/2858036.2858450>
- [14] Felix Kosmalla, Florian Daiber, Frederik Wiehr, and Antonio Krüger. 2017. ClimbVis: Investigating In-Situ Visualizations for Understanding Climbing Movements by Demonstration. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces* (Brighton, United Kingdom) (ISS '17). Association for Computing Machinery, New York, NY, USA, 270–279. <https://doi.org/10.1145/3132272.3134119>
- [15] Felix Kosmalla, Christian Murlowski, Florian Daiber, and Antonio Krüger. 2018. Slackliner - An Interactive Slackline Training Assistant. In *Proceedings of the 26th ACM International Conference on Multimedia* (Seoul, Republic of Korea) (MM '18). ACM, New York, NY, USA, 154–162. <https://doi.org/10.1145/3240508.3240537>
- [16] Beatriz Peres, Pedro F Campos, and Aida Azadegan. 2019. A Digitally-Augmented Ground Space with Timed Visual Cues for Facilitating Forearm Crutches' Mobility. In *IFIP Conference on Human-Computer Interaction*. Springer, 184–201.
- [17] David Anderson Richard A. Magill. 2016. *Motor Learning and Control: Concepts and Applications* (11th ed.). McGraw-Hill Education.
- [18] Sharmella Roopchand-Martin, Ricardo A Chong, Alison Facey, Pramanand Singh, and Akshai Mansing. 2018. A pilot randomised clinical trial comparing the effect of video game dance training with ladder drills on agility of elite volleyball players. *New Zealand Journal of Physiotherapy* 46, 4 (2018).
- [19] Yuji Sano, Koya Sato, Ryoichiro Shiraishi, and Mai Otsuki. 2016. Sports Support System: Augmented Ball Game for Filling Gap Between Player Skill Levels. In *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces* (Niagara Falls, Ontario, Canada) (ISS '16). ACM, New York, NY, USA, 361–366. <https://doi.org/10.1145/2992154.2996781>
- [20] Nina Schaffert and Klaus Mattes. 2015. Interactive Sonification in Rowing: Acoustic Feedback for On-Water Training. *IEEE MultiMedia* 22, 1 (2015), 58–67. <https://doi.org/10.1109/MMUL.2015.9>
- [21] Richard A Schmidt. 1991. Frequent augmented feedback can degrade learning: Evidence and interpretations. In *Tutorials in motor neuroscience*. Springer, 59–75.
- [22] Peter Schreiner. 2003. *Effective use of the agility ladder for soccer*. Reedswain Inc.
- [23] Yoonas A Sekhavat and Mohammad S Namani. 2018. Projection-based AR: Effective visual feedback in gait rehabilitation. *IEEE Transactions on Human-Machine Systems* 48, 6 (2018), 626–636.
- [24] Jeremy M Sheppard and Warren B Young. 2006. Agility literature review: Classifications, training and testing. *Journal of Sports Sciences* 24, 9 (2006), 919–932. <https://doi.org/10.1080/02640410500457109>
- [25] Rajinder Sodhi, Hrvoje Benko, and Andrew Wilson. 2012. LightGuide: projected visualizations for hand movement guidance. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 179–188.
- [26] Maurício Sousa, João Vieira, Daniel Medeiros, Artur Arsenio, and Joaquim Jorge. 2016. SleeveAR: Augmented Reality for Rehabilitation Using Realtime Feedback. In *Proceedings of the 21st International Conference on Intelligent User Interfaces* (Sonoma, California, USA) (IUI '16). ACM, New York, NY, USA, 175–185. <https://doi.org/10.1145/2856767.2856773>
- [27] Daniel Spelmezan, Mareike Jacobs, Anke Hilgers, and Jan Borchers. 2009. Tactile Motion Instructions for Physical Activities. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 2243–2252. <https://doi.org/10.1145/1518701.1519044>
- [28] Yuka Tanaka, Homare Kon, and Hideki Koike. 2018. A real-time golf-swing training system using sonification and sound image localization. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology*. 1–2.
- [29] Milka Trajkova and Francesco Cafaro. 2018. Takes Tutu to Ballet: Designing Visual and Verbal Feedback for Augmented Mirrors. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 1, Article 38 (March 2018), 30 pages. <https://doi.org/10.1145/3191770>
- [30] Frederik Wiehr, Felix Kosmalla, Florian Daiber, and Antonio Krüger. 2016. BetaCube: Enhancing Training for Climbing by a Self-Calibrating Camera-Projection Unit. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (CHI EA '16). Association for Computing Machinery, New York, NY, USA, 1998–2004. <https://doi.org/10.1145/2851581.2892393>
- [31] Warren B. Young, Brian Dawson, and Greg J. Henry. 2015. Agility and Change-of-Direction Speed are Independent Skills: Implications for Training for Agility in Invasion Sports. *International Journal of Sports Science & Coaching* 10, 1 (2015), 159–169. <https://doi.org/10.1260/1747-9541.10.1.159>